NDEJJE UNIVERSITY

Fear of God brings Knowledge and Wisdom

COURSE: COMPUTER SCIENCE

SEMESTER: II

COURSE UNIT: Cryptography and Cyber Security

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ASSIGNMENT: NETWORK SCANNING AND ANALYSIS USING NMAP

Introduction

Network security is an important aspect of modern computing systems. Organizations and individuals need to understand their networks to protect them from attacks. One of the most powerful tools used for network discovery and security auditing is Nmap (Network Mapper).

Task 1: Installation and Setup

Explain what Nmap is and its purpose.

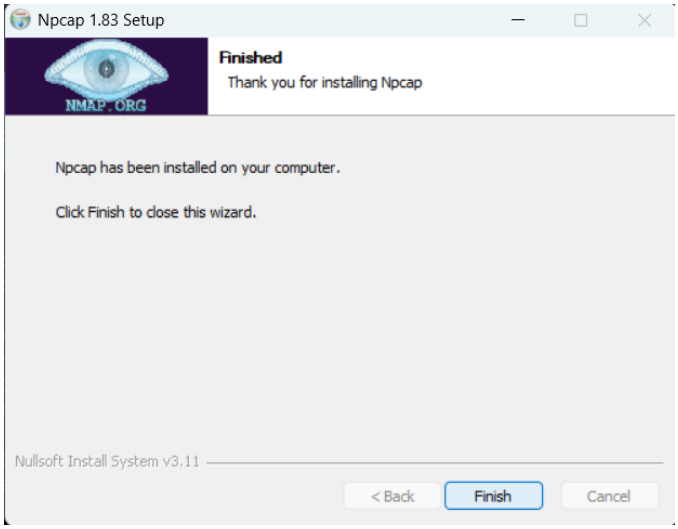
Nmap(Network Mapper) is a network scanning tool designed to discover hosts and services on a computer network. It works by sending specially crafted packets to target systems and analyzing their responses.

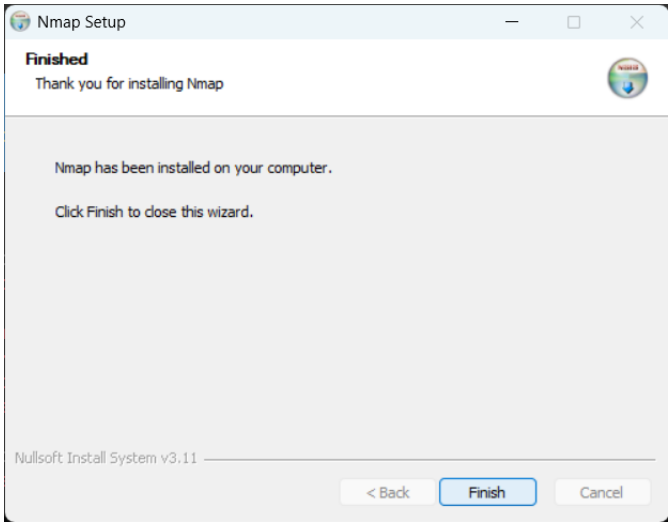
Main purposes of Nmap include:

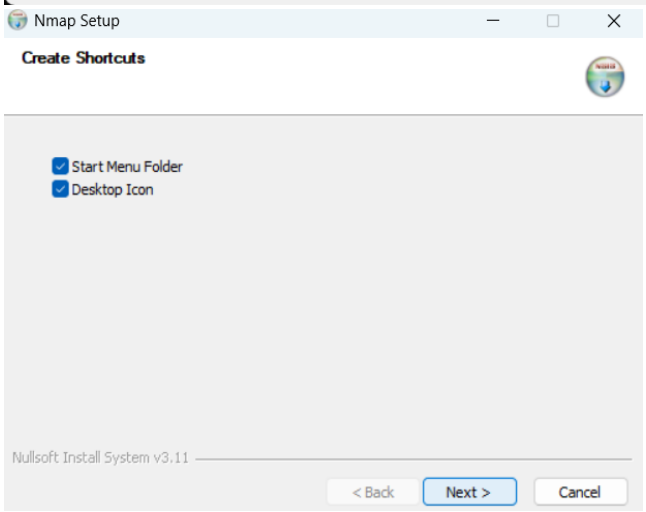
- **Host discovery** – identifying active devices on a network
- **Port scanning** – detecting open, closed, or filtered ports
- **Service detection** – identifying applications running on ports
- **Operating system detection** – determining the OS of a target machine
- **Security auditing** – identifying vulnerabilities and misconfigurations

Nmap is widely used in cybersecurity for:

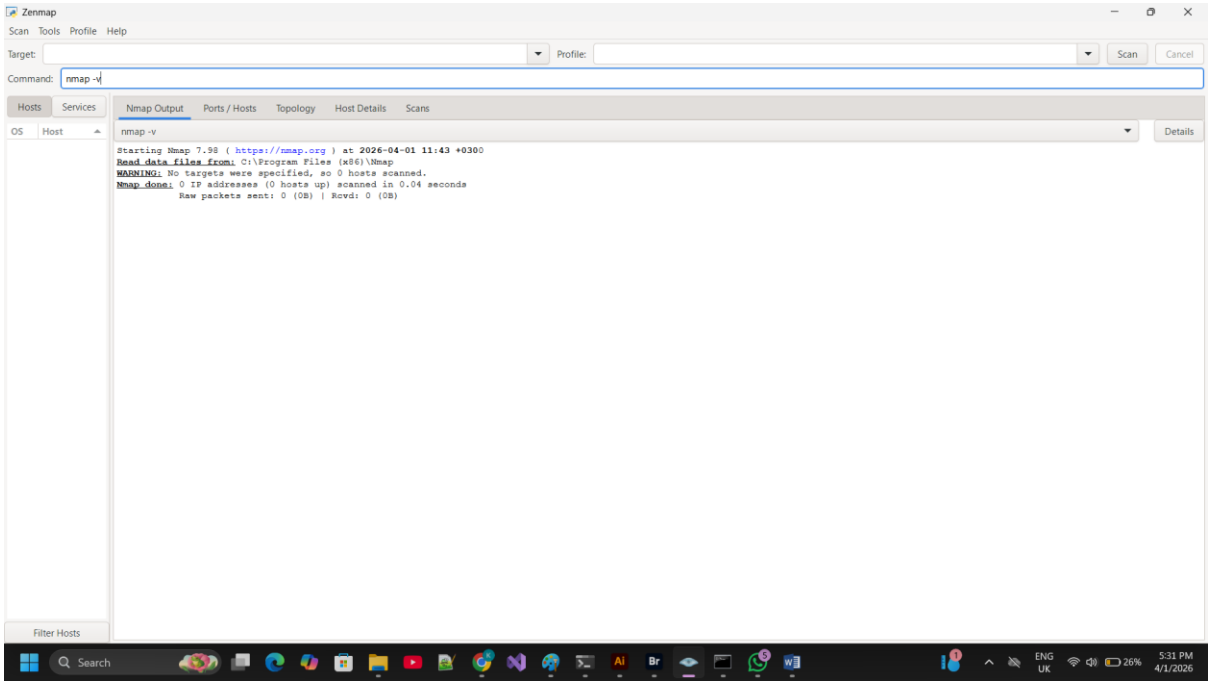
- Network inventory
- Monitoring system uptime
- Penetration testing
- Vulnerability assessment







Methodology: The command `nmap -v` was used to verify the installation. Analysis: Nmap was successfully installed. Nmap works by sending raw IP packets to a host and analyzing responses to create a map of the network.



Task 2: Host Discovery (15 Marks)

Methodology

The goal was to identify active devices on the subnet. The command nmap -sn 192.168.137.159/24 was used. The -sn flag (Ping Scan) disables port scanning to focus strictly on host availability using ICMP and ARP requests.

Results & Analysis

The scan discovered two active hosts on the 192.168.137.0/24 subnet:

192.168.137.1 (Umar.mshome.net): Identified as the network gateway.

192.168.137.159 (USER-EFQ6UCVS0S.mshome.net): Identified as the local workstation.

Analysis: The /24 notation instructed Nmap to scan 256 addresses. The speed of discovery (5.67 seconds) indicates a healthy local connection.

```
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\KIRABIRA KEITH>nmap -sn 192.168.1.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-01 11:29 +0300
Nmap done: 256 IP addresses (0 hosts up) scanned in 212.29 seconds

C:\Users\KIRABIRA KEITH>nmap -sn 192.168.137.159
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-01 11:34 +0300
Nmap scan report for USER-EFQ6UCVS0S.mshome.net (192.168.137.159)
Host is up.
Nmap done: 1 IP address (1 host up) scanned in 0.19 seconds

C:\Users\KIRABIRA KEITH>nmap -sn 192.168.137.159/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-01 11:35 +0300
Nmap scan report for Umar.mshome.net (192.168.137.1)
Host is up (0.023s latency).
MAC Address: E6:70:B8:0E:33:1E (Unknown)
Nmap scan report for USER-EFQ6UCVS0S.mshome.net (192.168.137.159)
Host is up.
Nmap done: 256 IP addresses (2 hosts up) scanned in 5.67 seconds

C:\Users\KIRABIRA KEITH>
```

Task 3: Port Scanning

Run 'nmap ' and 'nmap -p 1-1000 '. Identify open ports.

```
C:\Users\KIRABIRA KEITH>nmap 192.168.137.159
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-01 11:52 +0300
Nmap scan report for USER-EFQ6UCVS0S.mshome.net (192.168.137.159)
Host is up (0.00098s latency).
Not shown: 988 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
80/tcp   open  http
135/tcp  open  msrpc
139/tcp  open  netbios-ssn
443/tcp  open  https
445/tcp  open  microsoft-ds
902/tcp  open  iss-realsecure
912/tcp  open  apex-mesh
2869/tcp open  icslap
3306/tcp open  mysql
6666/tcp open  irc
8080/tcp open  http-proxy

Nmap done: 1 IP address (1 host up) scanned in 3.07 seconds
```

```
C:\Users\KIRABIRA KEITH>nmap -p 1-1000 192.168.137.159
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-01 11:50 +0300
Nmap scan report for USER-EFQ6UCVS0S.mshome.net (192.168.137.159)
Host is up (0.00038s latency).
Not shown: 991 closed tcp ports (reset)
PORT    STATE    SERVICE
21/tcp  open     ftp
80/tcp  open     http
135/tcp open     msrpc
137/tcp filtered netbios-ns
139/tcp open     netbios-ssn
443/tcp open     https
445/tcp open     microsoft-ds
902/tcp open     iss-realsecure
912/tcp open     apex-mesh

Nmap done: 1 IP address (1 host up) scanned in 4.22 seconds
```

Methodology

Two commands were used: `nmap` 192.168.137.159 and `nmap` -p 1-1000 192.168.137.159.

Two variations of port scanning were performed:

nmap 192.168.137.159: Scanned the top 1,000 most common ports.

nmap -p 1-1000 192.168.137.159: Explicitly scanned every port from 1 to 1,000.

Results

The scan identified 12 open ports. Key findings include:

Port 80/443 (HTTP/HTTPS): Standard web services.

Port 3306 (MySQL): Database services (Maria DB).

Port 21 (FTP): File transfer services.

Analysis: The -p 1-1000 scan revealed that port 137 was "filtered," meaning a firewall (Windows Defender) was actively dropping packets for that specific service.

Task 4: Service Detection

Run 'nmap -sV ' and identify running services.

```
C:\Users\KIRABIRA KEITH>nmap -sV 192.168.137.159
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-01 11:53 +0300
Nmap scan report for USER-EFQ6UCVS0S.mshome.net (192.168.137.159)
Host is up (0.000026s latency).
Not shown: 988 closed tcp ports (reset)
PORT     STATE SERVICE         VERSION
21/tcp   open  ftp             FileZilla ftpd 0.9.41 beta
80/tcp   open  http            Apache httpd 2.4.58 ((Win64) OpenSSL/3.1.3 PHP/8.0.30)
135/tcp  open  msrpc           Microsoft Windows RPC
139/tcp  open  netbios-ssn     Microsoft Windows netbios-ssn
443/tcp  open  ssl/http        Apache httpd 2.4.58 ((Win64) OpenSSL/3.1.3 PHP/8.0.30)
445/tcp  open  microsoft-ds?
902/tcp  open  ssl/vmware-auth VMware Authentication Daemon 1.10 (Uses VNC, SOAP)
912/tcp  open  vmware-auth     VMware Authentication Daemon 1.0 (Uses VNC, SOAP)
2869/tcp open  http            Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
3306/tcp open  mysql           MariaDB 10.3.23 or earlier (unauthorized)
6666/tcp open  irc?
8080/tcp open  http-proxy?
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 118.30 seconds
```

Methodology

The command nmap -sV 192.168.137.159 was used to determine the specific software versions running on the open ports.

Results & Analysis

Web Server: Apache httpd 2.4.58 (running PHP 8.0.30).

Database: MariaDB 10.3.23 (found on port 3306).

FTP Server: FileZilla ftpd 0.9.41 beta.

Analysis: Identifying these versions is critical. For example, knowing the exact Apache version allows an administrator to check if the server is vulnerable to known exploits.

Task 5: OS Detection

Run 'nmap -O ' and identify the operating system

```
C:\Users\KIRABIRA KEITH>nmap -O 192.168.137.159
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-01 11:57 +0300
Nmap scan report for USER-EFQ6UCVS0S.mshome.net (192.168.137.159)
Host is up (0.00051s latency).
Not shown: 990 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
80/tcp   open  http
135/tcp  open  msrpc
139/tcp  open  netbios-ssn
443/tcp  open  https
445/tcp  open  microsoft-ds
902/tcp  open  iss-realsecure
912/tcp  open  apex-mesh
2869/tcp open  icslap
3306/tcp open  mysql
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.98%E=4%D=4/1%OT=21%CT=1%CU=41132%PV=Y%DS=0%DC=L%G=Y%TM=69CCDE01
OS:%P=i686-pc-windows-windows)SEQ(SP=101%GCD=1%ISR=109%TI=I%CI=I%II=I%SS=S%
OS:TS=A)SEQ(SP=104%GCD=1%ISR=10E%TI=I%CI=I%II=I%SS=S%TS=A)SEQ(SP=105%GCD=1%
OS:ISR=10E%TI=I%CI=I%II=I%SS=S%TS=A)SEQ(SP=107%GCD=4%ISR=10A%TI=I%CI=I%II=I
OS:%SS=S%TS=A)SEQ(SP=FC%GCD=1%ISR=109%TI=I%CI=I%II=I%SS=S%TS=A)OPS(O1=MFFD7
OS:NW8ST11%O2=MFFD7NW8ST11%O3=MFFD7NW8NNT11%O4=MFFD7NW8ST11%O5=MFFD7NW8ST11
OS:%O6=MFFD7ST11)WIN(W1=FFFF%W2=FFFF%W3=FFFF%W4=FFFF%W5=FFFF%W6=FFFF)ECN(R=
OS:Y%DF=Y%T=80%W=FFFF%O=MFFD7NW8NNS%CC=N%Q=)T1(R=Y%DF=Y%T=80%S=O%A=S+%F=AS%
OS:RD=0%Q=)T2(R=N)T3(R=N)T4(R=Y%DF=Y%T=80%W=0%S=A%A=O%F=R%O=%RD=0%Q=)T5(R=Y
OS:%DF=Y%T=80%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)T6(R=Y%DF=Y%T=80%W=0%S=A%A=O%F=R
OS:%O=%RD=0%Q=)T7(R=N)U1(R=Y%DF=N%T=80%IPL=164%UN=0%RIPL=G%RID=G%RIPCK=G%RU
OS:CK=G%RUD=G)IE(R=Y%DFI=N%T=80%CD=Z)

Network Distance: 0 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 15.12 seconds

C:\Users\KIRABIRA KEITH>nmap -0 172.17.31.209
nmap: illegal option -- 0
See the output of nmap -h for a summary of options.

C:\Users\KIRABIRA KEITH>
```

Methodology

The command nmap -O 192.168.137.159 was executed to perform TCP/IP fingerprinting to guess the Operating System.

Results & Analysis

Nmap provided a detailed TCP/IP fingerprint. While an exact build of Windows wasn't "matched" with 100% certainty, the Service Info correctly identified the family: OS: Windows; CPE: cpe:/o:microsoft:windows.

Task 6: Advanced Scans

Run advanced scans such as SYN (-sS), UDP (-sU), and Aggressive (-A). Compare results.

nmap -sS 192.168.137.159

```
C:\Users\KIRABIRA KEITH>nmap -sS 192.168.137.159
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-01 18:30 +0300
Nmap scan report for USER-EFQ6UCVS0S.mshome.net (192.168.137.159)
Host is up (0.000015s latency).
Not shown: 990 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
80/tcp   open  http
135/tcp  open  msrpc
139/tcp  open  netbios-ssn
443/tcp  open  https
445/tcp  open  microsoft-ds
902/tcp  open  iss-realsecure
912/tcp  open  apex-mesh
2869/tcp open  icslap
3306/tcp open  mysql

Nmap done: 1 IP address (1 host up) scanned in 3.39 seconds
```

nmap -sU 192.168.137.159

```
C:\Users\KIRABIRA KEITH>nmap -sU 192.168.137.159
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-01 18:35 +0300
Nmap scan report for USER-EFQ6UCVS0S.mshome.net (192.168.137.159)
Host is up (0.00019s latency).
Not shown: 991 closed udp ports (port-unreach)
PORT     STATE         SERVICE
123/udp  open|filtered ntp
137/udp  open|filtered netbios-ns
138/udp  open|filtered netbios-dgm
500/udp  open|filtered isakmp
1900/udp open          upnp
4500/udp open|filtered nat-t-ike
5050/udp open|filtered mmcc
5353/udp open|filtered zeroconf
5355/udp open|filtered llmnr

Nmap done: 1 IP address (1 host up) scanned in 178.24 seconds
```

nmap -A 192.168.137.159

```
C:\Users\KIRABIRA KEITH>nmap -A 192.168.137.159
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-01 18:47 +0300
Nmap scan report for USER-EFQ6UCVS0S.mshome.net (192.168.137.159)
Host is up (0.00046s latency).
Not shown: 990 closed tcp ports (reset)
PORT     STATE SERVICE         VERSION
21/tcp   open  ftp             FileZilla ftpd 0.9.41 beta
| ftp-syst:
|_  SYST: UNIX emulated by FileZilla
80/tcp   open  http            Apache httpd 2.4.58 ((Win64) OpenSSL/3.1.3 PHP/8.0.30)
|_http-server-header: Apache/2.4.58 (Win64) OpenSSL/3.1.3 PHP/8.0.30
| http-title: Welcome to XAMPP
|_Requested resource was http://USER-EFQ6UCVS0S.mshome.net/dashboard/
135/tcp  open  msrpc           Microsoft Windows RPC
139/tcp  open  netbios-ssn     Microsoft Windows netbios-ssn
443/tcp  open  ssl/http        Apache httpd 2.4.58 ((Win64) OpenSSL/3.1.3 PHP/8.0.30)
|_http-server-header: Apache/2.4.58 (Win64) OpenSSL/3.1.3 PHP/8.0.30
|_ssl-date: TLS randomness does not represent time
| ssl-cert: Subject: commonName=localhost
| Not valid before: 2009-11-10T23:48:47
|_Not valid after:  2019-11-08T23:48:47
| tls-alpn:
|_  http/1.1
| http-title: Welcome to XAMPP
|_Requested resource was https://USER-EFQ6UCVS0S.mshome.net/dashboard/
445/tcp  open  microsoft-ds?
902/tcp  open  ssl/vmware-auth VMware Authentication Daemon 1.10 (Uses VNC, SOAP)
912/tcp  open  vmware-auth     VMware Authentication Daemon 1.0 (Uses VNC, SOAP)
2869/tcp open  http            Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
3306/tcp open  mysql           MariaDB 10.3.23 or earlier (unauthorized)
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.98%E=4%D=4/1%OT=21%CT=1%CU=36860%PV=Y%DS=0%DC=L%G=Y%TM=69CD3E40
OS:%P=i686-pc-windows-windows)SEQ(SP=100%GCD=2%ISR=105%TI=I%CI=I%II=I%SS=S%
OS:TS=A)SEQ(SP=104%GCD=1%ISR=10D%TI=I%CI=I%II=I%SS=S%TS=A)SEQ(SP=108%GCD=1%
OS:ISR=108%TI=I%CI=I%II=I%SS=S%TS=A)SEQ(SP=F7%GCD=1%ISR=107%TI=I%CI=I%II=I%
OS:SS=S%TS=A)SEQ(SP=FE%GCD=3%ISR=10E%TI=I%CI=I%II=I%SS=S%TS=A)OPS(O1=MFFD7N
OS:W8ST11%O2=MFFD7NW8ST11%O3=MFFD7NW8NNT11%O4=MFFD7NW8ST11%O5=MFFD7NW8ST11%
OS:O6=MFFD7ST11)WIN(W1=FFFF%W2=FFFF%W3=FFFF%W4=FFFF%W5=FFFF%W6=FFFF)ECN(R=Y
OS:%DF=Y%T=80%W=FFFF%O=MFFD7NW8NNS%CC=N%Q=)T1(R=Y%DF=Y%T=80%S=O%A=S+%F=AS%R
OS:D=0%Q=)T2(R=Y%DF=Y%T=80%W=0%S=Z%A=S%F=AR%O=%RD=0%Q=)T3(R=Y%DF=Y%T=80%W=0
OS:%S=Z%A=O%F=AR%O=%RD=0%Q=)T4(R=Y%DF=Y%T=80%W=0%S=A%A=O%F=R%O=%RD=0%Q=)T5(
OS:R=Y%DF=Y%T=80%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)T6(R=Y%DF=Y%T=80%W=0%S=A%A=O%
OS:F=R%O=%RD=0%Q=)T7(R=Y%DF=Y%T=80%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)U1(R=Y%DF=N
OS:%T=80%IPL=164%UN=0%RIPL=G%RID=G%RIPCK=G%RUCK=G%RUD=G)IE(R=Y%DFI=N%T=80%C
OS:D=Z)

Network Distance: 0 hops
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
| smb2-time:
|   date: 2026-04-01T15:48:05
|_  start_date: N/A
| smb2-security-mode:
|   3.1.1:
|_    Message signing enabled and required

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 43.59 seconds

C:\Users\KIRABIRA KEITH>s
```

Methodology

Three types of scans were compared:

1. SYN Scan (-sS): Fast and "half-open," as it doesn't complete the TCP handshake.
2. UDP Scan (-sU): Discovered services like NTP (123) and UPnP (1900).
3. Aggressive Scan (-A): Combined version detection, OS detection, and traceroute.

Analysis: The UDP scan is essential because services like DNS and DHCP do not use TCP. The Aggressive scan revealed that SMBv3 signing is enabled on the host, which is a positive security configuration.

Comparison Table

Scan Type	Command Flag	Key Observation
SYN Stealth	`-sS`	Fast (3.39s). Identified 10 TCP ports without full connection.
UDP Scan	`-sU`	Slow (178s). Found NTP (123) and UPnP (1900) which TCP missed.
Aggressive	`-A`	Comprehensive. Combined OS, Version, and SMB security info.

Task 7: Vulnerability Scan

Run 'nmap --script vuln ' and identify vulnerabilities.

```
C:\Users\KIRABIRA KEITH>nmap --script vuln 192.168.137.159
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-01 18:56 +0300
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 16.91 seconds

C:\Users\KIRABIRA KEITH>
```

Methodology:

The command `nmap` --script vuln 192.168.137.159 was used.

Analysis: The host initially appeared "down" during this scan. This is a common security response where the Windows Firewall detects a high volume of script probes and temporarily blocks the scanning IP to prevent a potential attack.

Task 8: Other Essential Nmap Commands

Command: `nmap -T4 -F 192.168.137.159`

```
:\Users\KIRABIRA KEITH>nmap -T4 -F 192.168.137.159
tarting Nmap 7.98 ( https://nmap.org ) at 2026-04-02 17:13 +0300
ote: Host seems down. If it is really up, but blocking our ping probes, try -Pn
map done: 1 IP address (0 hosts up) scanned in 7.25 seconds

:\Users\KIRABIRA KEITH>
```

Speeds up the execution by adjusting parallel probes and timeouts. `-F` limits the scan to the top 100 most common ports instead of 1,000.

Command: `nmap` -oN nmap_report.txt 192.168.137.159

```
C:\Users\KIRABIRA KEITH>nmap -oN nmap_report.txt 192.168.137.159
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-02 17:20 +0300
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 11.69 seconds

C:\Users\KIRABIRA KEITH>
```

The -oN flag saves the output in "Normal" format into a file named `nmap_report.txt` in your current folder.

Command: nmap -Pn 192.168.137.159

```
C:\Users\KIRABIRA KEITH>nmap -Pn 192.168.137.159
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-02 17:39 +0300
Nmap scan report for 192.168.137.159
Host is up.
All 1000 scanned ports on 192.168.137.159 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 209.70 seconds

C:\Users\KIRABIRA KEITH>
```

The -Pn flag tells Nmap to skip the initial "ping" check and assume the host is active. This is exactly what was needed when task 7 scan initially failed.

Conclusion

The assignment helped me to use Nmap for network reconnaissance. By discovering open ports and identifying service versions, I learned how to assess a machine's security posture. The discovery of an open database and web server highlights the importance of using firewalls to restrict access to only authorized users.